

Required Components

Hardware (Simulated)

Component	Quantity
Raspberry Pi Pico W	1
HC-SR04 Ultrasonic Sensor	1
Breadboard	1
Jumper Wires	As Required
LED (Optional)	1
Buzzer (Optional)	1

Software

Software	Purpose
Wokwi	Circuit Simulation
MicroPython	Programming
Thonny IDE	Code Development
ThingSpeak	IoT Dashboard
GitHub	Project Repository

Internet Requirement

Wi-Fi connection for sending data to ThingSpeak.